

ConX Orbit

Database Schema & Local Data Storage Guide

Admin CMS · Solutions · Case Studies · Team · Assets

This document describes the data model used by the ConX Orbit admin portal and public website API. Use it to recreate the same schema in MongoDB, PostgreSQL, MySQL, or any other database.

1. Storage overview (local)

Database file: backend/data/db.json (LowDB JSON store)

Asset files (disk): backend/uploads/

Public asset URL: http://localhost:5000/uploads/...

Images are stored as files on disk. The database only stores **string paths/URLs** (e.g. /uploads/solutions/panel-x.jpg), never binary blobs.

Collections / tables: solutions · caseStudies · team · admin

Seed command: cd backend && npm run seed

2. Table: solutions

One document/row per product or workflow solution. Drives Solutions listing, solution detail pages, Portfolio sections (via portfolioCategory), and trade pages (via trades[]).

Field	Type	Notes
_id	string	Primary key
slug	string (unique)	URL key, e.g. panel-x
name	string	Display name
description	string	Listing card text
listingBadge	string	e.g. Client System / Internal Product
categories	string[]	Tags / filters
image	string (URL)	Listing image path under /uploads
portfolioImage	string (URL)	Portfolio-specific image (optional)
portfolioCategory	enum	client_system internal_product workflow_solution none
trades	string[]	e.g. ["facade"] — which trade pages show this
sortOrder	number	Sort order
published	boolean	Visible if true
showOnListing	boolean	Show on /solutions page
confidential	boolean	Confidential portfolio item
confidentialLabel	string	Label text when confidential
caseStudySlug	string	Linked case study slug for CTA
portfolioStatus	string	e.g. In Development / Capability

portfolioLabel	string	Portfolio section label
portfolioBody	string	Portfolio description override
seo.title	string	SEO title (leave empty for now)
seo.description	string	SEO description (empty for now)
seo.ogImage	string	OG image URL (empty for now)
detail	object (JSON)	Full solution page content (see below)
createdAt	datetime	Created timestamp
updatedAt	datetime	Updated timestamp

2.1 Nested object: detail

Stored as JSON/object on the solution. Section order on the public page is fixed in code; only field values change. Section badges (Capability Showcase, etc.) stay hardcoded in UI.

```

detail
■■■ title / titleBefore / titleHighlight / titleAfter
■■■ body
■■■ primaryCta { label, href }
■■■ demoCta { label, targetId }
■■■ heroImage           ← asset URL
■■■ stats { bestFor, coreFunction, platform, workflow }
■■■ challenge { title, body, problems[{ title, body }] }
■■■ capabilities { title, body, cards[{ title, body, icon? }] }
■■■ howItWorks { title, stages[{ title, body }] }
■■■ builtFor { title, body, audiences[{ title, body, icon? }] }
■■■ demo { title, videoSrc, posterSrc } ← posterSrc asset URL
■■■ more { title, slugs[] }
■■■ faq { titleBefore, titleHighlight, titleAfter, items[{ question, answer }] }
■■■ cta { title, body, primary{ label, href }, secondary? }

```

3. Table: caseStudies

Field	Type	Notes
_id	string	Primary key
slug	string (unique)	e.g. facade-panel-visibility
title	string	Case study title
industry	string	Industry label
summary	string	Short summary
problem	string	Problem section
built	string	What we built
result	string	Result section
quote.text	string	Client quote
quote.author	string	Quote author
sortOrder	number	Sort order
published	boolean	Visible if true
seo.title	string	Empty for now
seo.description	string	Empty for now
seo.ogImage	string	Empty for now
createdAt / updatedAt	datetime	Timestamps

4. Table: team (team members)

Field	Type	Notes
_id	string	Primary key
name	string	Full name
designation	string	Designation
role	string	Role shown on About page
socialLinks	array	[[platform, url]] e.g. linkedin
image	string (URL)	Photo path under /uploads
quote	string	Quote text
quoteAuthor	string	Quote attribution
sortOrder	number	Sort order
published	boolean	Visible if true
createdAt / updatedAt	datetime	Timestamps

5. Table: admin

Field	Type	Notes
email	string	Admin login email
passwordHash	string	bcrypt hash (never store plain password)

Default after seed: admin@conxorbit.com / admin123 (change in production).

6. Assets

Disk folder: backend/uploads/ (examples: uploads/solutions/panel-x.jpg)

In DB: only string path, e.g. /uploads/solutions/panel-x.jpg

In browser: API base + path → http://localhost:5000/uploads/solutions/panel-x.jpg

Solution fields that hold asset links: **image**, **portfolioImage**, **detail.heroImage**, **detail.demo.posterSrc**, **seo.ogImage**.

7. Seeded counts (after npm run seed)

- solutions — ~10 (9 public + 1 confidential portfolio-only)
- caseStudies — 3
- team — 3
- admin — 1

8. How pages use the data

- **/solutions** — published + showOnListing solutions
- **/solutions/:slug** — full detail object
- **/portfolio** — split by portfolioCategory; case studies for modal
- **/case-studies/facade** related cards — solutions where trades includes "facade"
- **/about** — published team members
- **/admin** — CRUD for all of the above